

BALASORE, IndiaWMO: **428950**Lat: **21.517N**Lon: **86.933E**Elev: **20**StdP: **101.08**Time Zone: **5.50 (IND)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.2	14.3	5.8	5.7	22.2	7.3	6.4	20.2	4.2	21.5	3.4	21.4	0.9	0	0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.6	37.6	27.0	36.2	27.0	35.1	27.0	29.4	34.3	28.9	33.8	28.5	33.2	2.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.2	24.4	32.9	27.7	23.7	32.4	27.2	23.0	31.8	97.6	34.6	95.1	33.9	92.7	33.3	34.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.7	3.5	3.2	10.9	40.9	1.0	2.4	10.2	42.6	9.6	44.0	9.1	45.4	8.3	47.2		
(5)			DB													
			WB	8.6	30.7	0.9	1.3	8.0	31.7	7.5	32.4	7.0	33.1	6.4	34.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	20.8	24.0	27.9	30.2	30.8	30.3	29.3	29.1	28.9	27.8	24.7	21.3	(6)	
	DBStd	3.78	1.78	2.18	1.87	2.11	1.85	1.91	1.24	1.11	1.30	1.52	1.73	1.70	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	3	2	0	0	0	0	0	0	0	0	0	0	1	(9)	
	CDD10.0	6248	335	393	556	607	646	609	598	592	568	552	440	351	(10)	
	CDD18.3	3208	78	160	298	357	387	359	340	334	318	293	190	93	(11)	
	CDH23.3	36709	407	1147	3181	4642	5363	4830	4221	4078	3816	3127	1403	495	(12)	
	CDH26.7	16342	59	367	1393	2460	3004	2487	1786	1644	1509	1151	413	69	(13)	
(14) Wind	WSAvg	1.1	0.9	1.0	1.3	1.7	1.6	1.3	1.2	1.1	0.9	0.8	0.8	0.9	(14)	
(15) Precipitation	PrecAvg	1633	22	30	27	60	131	270	302	300	285	169	31	6	(15)	
	PrecMax	2153	106	153	185	138	401	606	564	543	567	668	225	44	(16)	
	PrecMin	1128	2	0	0	0	12	100	124	120	43	0	0	0	(17)	
	PrecStd	285	29	42	38	45	79	104	99	95	130	160	45	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.0	34.2	36.9	39.9	39.6	39.0	35.2	34.2	34.3	34.1	32.1	29.8	(19)	
		MCWB	18.9	21.3	23.6	25.9	26.9	28.4	28.3	27.9	27.5	26.2	22.8	20.1	(20)	
	2%	DB	28.1	32.1	35.1	37.6	37.3	36.5	33.8	33.2	33.5	33.1	31.0	28.4	(21)	
		MCWB	18.7	21.1	23.4	26.1	28.0	28.6	27.9	27.7	27.4	26.1	22.8	19.4	(22)	
	5%	DB	27.0	30.5	33.8	36.1	35.9	35.1	32.7	32.4	32.7	32.3	30.1	27.3	(23)	
		MCWB	18.1	20.7	23.3	26.3	27.9	28.4	27.5	27.5	27.2	25.8	22.3	18.7	(24)	
	10%	DB	25.7	29.1	32.6	34.7	34.7	34.0	31.9	31.6	31.9	31.4	29.1	26.2	(25)	
		MCWB	17.7	20.6	23.3	26.2	27.9	28.0	27.3	27.2	27.0	25.4	21.7	18.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.1	24.9	27.1	28.9	30.1	30.1	29.0	28.7	28.6	27.7	25.6	22.4	(27)	
		MCDB	26.3	29.7	32.9	35.0	35.1	35.0	33.2	32.5	32.6	32.2	29.7	26.8	(28)	
	2%	WB	20.9	23.8	26.3	28.2	29.5	29.5	28.4	28.2	28.0	27.2	24.6	21.2	(29)	
		MCDB	25.0	28.3	31.6	34.3	34.5	34.3	32.4	32.0	32.1	31.4	28.5	25.7	(30)	
	5%	WB	20.1	23.0	25.8	27.7	29.1	28.9	27.9	27.7	27.6	26.7	24.0	20.3	(31)	
		MCDB	24.5	27.2	30.6	33.6	34.1	33.7	31.8	31.4	31.4	30.7	28.0	25.1	(32)	
	10%	WB	19.3	22.3	25.2	27.2	28.6	28.3	27.5	27.5	27.2	26.3	23.2	19.5	(33)	
		MCDB	23.6	26.4	29.8	32.8	33.4	33.0	31.1	30.9	30.8	30.0	27.0	24.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	10.5	9.9	9.6	8.6	6.5	5.0	4.9	5.3	6.9	9.5	10.9	(35)	
		MCDBR	12.2	11.9	11.0	11.0	9.9	8.6	6.5	6.1	6.6	7.9	9.8	11.4	(36)	
	5% WB	MCWBR	4.9	4.5	4.2	4.2	4.5	3.9	2.8	2.6	2.7	3.1	3.9	4.8	(37)	
		MCWBR	9.9	9.4	9.5	9.5	8.7	8.1	5.8	5.5	6.0	6.6	7.6	9.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.575	0.557	0.610	0.624	0.666	0.686	0.594	0.572	0.567	0.580	0.612	0.619	(40)	
		taud	1.764	1.843	1.773	1.795	1.745	1.713	1.925	1.999	1.998	1.905	1.753	1.700	(41)	
	Edn at Noon		687	741	723	720	686	666	731	750	747	713	651	629	(42)	
		Edh at Noon	207	202	225	223	233	239	193	180	177	187	208	214	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.10	4.95	5.74	6.30	6.15	4.69	4.04	4.22	4.25	4.47	4.21	3.81	(44)	
	RadStd		0.23	0.30	0.27	0.26	0.31	0.36	0.35	0.28	0.25	0.41	0.25	0.31	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	+0.33	+0.38	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	+0.40	+0.33	N/A	N/A	N/A	N/A

Nomenclature: See separate page